

KOH WAN LENG TRINITY

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OBJECTIVE

Year 3 Aerospace Engineering student from Republic Polytechnic, drawn to aircraft structures, aerodynamics, and simulation-based design, with aspirations toward design and R&D. Hands-on experience spans CubeSat payload systems, flight training, and national-level STEM leadership as Vice President of the Institute of Physics Singapore Student Chapter (IPSSC), a university-led physics body in which I am the only pre-university representative. Looking to bring rigorous analytical thinking and a genuine love of flight physics to an aerospace engineering internship.

EDUCATION

Republic Polytechnic

Diploma in Aerospace Engineering

2023 – 2027 (Expected)

Relevant Modules:

Aerodynamics & Propulsion, Thermofluids, Principles of Mechanics, Engineering Mathematics, Engineering Materials, Aircraft Inspection, Aviation Maintenance Practices, Digital Techniques & Electronic Instrument Systems, Electrical & Electronic Fundamentals, Fundamentals of Industrial Internet of Things, Unmanned Aircraft Operations, Programming and Data Analysis

Evergreen Secondary School

GCE O-Level Certificate, 6 Credits

2019 – 2022

Woodgrove Primary School

PSLE Certificate

2013 – 2018

KEY SKILLS

Technical Skills • 3D modelling & parametric design • Printed circuit board (PCB) design • Satellite signal processing, SDR & GMSK modulation • Raspberry Pi CM4 payload integration • Pixhawk flight controller integration • Arduino microcontroller programming • Drone assembly & testing • Wire locking & wire crimping • Soldering • Electrical circuit assembly • Hovercraft construction • Diamond DA-40 flight training (SYFC BFC Phase 1) • Aerodynamic, propulsion & thermofluid systems analysis • Orbital/mission simulation & analysis	Computer, Software & Programming • Microsoft Office (Word, Excel, PowerPoint) • Autodesk Fusion 360 • KiCad • GNURadio • Blynk IoT cloud platform • NASA GMAT • Tableau • LTSpice • Arduino IDE • Canva • C++ (basic) • Python (basic) • VBA / Excel
Languages • English (written & spoken) • Chinese (written & spoken)	Core Skills • Multi-institution event leadership (PLANCKS, IAPS) • Committee & subcommittee management • Public science communication & STEM outreach • Publicity leadership & graphic design • Cross-team & external stakeholder liaison • First Aid, CPR & AED certified

WORK EXPERIENCE

Space Faculty <i>Intern (Mar – Apr 2024) Freelance Instructor, SpaceBuds (Mar – Apr 2025)</i> • Taught space-themed lessons for SpaceBuds, Space Faculty's kindergarten space-education programme. • Proposed, designed and built the SpaceBuds Question Board, an interactive tool sparking children's curiosity about space. • Facilitated Integrated Space Programme (ISP) Phase 1, a 1-week build of a fully functional (non-flight-rated) satellite from scratch. • Publicised partnerships and competitions. • Researched legal and regulatory requirements for sending biological seed samples to space for a seed-to-space launch.	2024 – 2025
Swimwerks Asia Pte. Ltd. <i>Swimming Teacher & Lifeguard</i> • AustSwim-licensed instructor teaching children swimming skills across ability levels. • Lifeguard duty: pool surveillance, CPR/AED/First Aid emergency response.	2022 – 2024
Kumon Asia & Oceania Pte. Ltd. <i>Mathematics & English Instructor</i> • Taught Mathematics and English to primary-level students using the Kumon method, with progress tracking.	2019

LEADERSHIP & KEY PROJECTS

Final Year Project, Cloud IoT-Enabled MRO Drone

Apr 2026 – Present

Republic Polytechnic

- Developing a Pixhawk-based MRO inspection drone with cloud IoT connectivity as my capstone project; writing onboard firmware and designing the IoT data pipeline to a cloud monitoring dashboard.

Defence Science and Technology Agency (DSTA) BrainHack 2025, CubeSat Challenge

11–12 Jun 2025

Participant

- Built a functional satellite prototype under hackathon time constraints.

NASA Mars Food Design Challenge, SGAC Space Exploration Project Group

May 2026 – Present

Project Lead

- Leading a ~50-person, multidisciplinary, multi-country project team (spanning varying educational levels) under the Space Generation Advisory Council, designing food systems for crewed Mars missions.

Integrated Space Programme (ISP) Phase 2, Space Faculty x NuSpace

2024 – 2026

Payload Team & Marketing Crew, Project MangoSat

- MangoSat is a 3U CubeSat (~4kg) built for multispectral Earth observation, monitoring NDVI (vegetation health) across Singapore's forests, southern islands and shipping routes; presented at the project's Critical Design Review (Mar 2025).
- Utilized NASA's General Mission Analysis Tool (GMAT) for orbital and mission simulations, analyzing various launch vehicles, target altitudes, and inclinations.
- Contributed to the software environment for the Raspberry Pi CM4 payload and assisted with GNURadio/SDR testing for satellite downlink communications.
- Presented MangoSat's mission objectives and payload architecture to industry professionals and public audiences at the Singapore Space Summit and Singapore Science and Engineering Fair (SSEF).
- Designed the official project mascot, stickers, and promotional Instagram Reels to drive public engagement and STEM outreach.

School of Engineering (SEG) Overseas Learning Journey, Bangkok, Thailand

8–14 Sep 2024

Republic Polytechnic, School of Engineering

- Technical visits to MRO and aerospace facilities (airport line maintenance operations, aviation repair and component overhaul shops), alongside cultural immersion at the Grand Palace and Royal Navy Museum.

Institute of Physics Singapore Student Chapter (IPSSC)

2024 – Present

Vice President (AY25/26) | Publicity Manager (AY24/25)

- VP overseeing a 28-member national committee spanning NUS, NTU and SUTD - the only pre-university representative in this university-led body - supporting all subcommittees (publicity, logistics, events) on flagship events including PLANCKS 2026.
- Organised and publicised PLANCKS 2026: a 2-day, multi-venue competition at NTU featuring 13 teams, guest talks and a panel discussion with researchers from CQT, ST Engineering, Atomionics and BiQut.
- Publicity Manager (AY24/25): led a 6-person team within the 20-member committee, driving outreach for PLANCKS 2025 (3-day event, NUS professor guest talks, EdiPhy panel).
- Ran Street Astronomy outreach at Wisma Geylang Serai, designing Artemis-mission posters and engaging the public to spark interest in STEM and space.

Singapore Youth Flying Club (SYFC), Basic Flying Course (BFC) 289

2024 – 2025

Course In Charge

- Course In Charge: primary liaison between SYFC flight operations staff and the BFC 289 trainee cohort - attendance tracking, scheduling support, day-to-day admin.
- Designed BFC 289's official course patch, from digital artwork through embroidered production.
- Completed Basic Flying Course Phase 1, flying supervised sorties on the Diamond DA-40 under CAAS-regulated training.

Space Faculty International Space Challenge (ISC)

2024, 2025, 2026

Participant, 3 consecutive years

CO-CURRICULAR ACTIVITIES, REPUBLIC POLYTECHNIC

Student Council, Branding Sub-Committee	2023 – 2024
Created Instagram/TikTok content for mid-semester and festive campaigns; Student Ambassador for RP Open House.	
School of Engineering (SEG) Club	2023 – 2024
Publicity In Charge for SEG Bonding Day; First Aider at the Student Leadership Preparatory Camp.	
RP Astronomy Club & Galaxy Astronomy Club (Woodlands Galaxy CC)	2023 – 2024
Lyrids Star Party, Desaru and Pulau Ubin stargazing camps; telescope operation.	

EXTERNAL ENRICHMENT PROGRAMMES & VOLUNTEERING

Singapore Airshow 2026	Feb 2026
Attended Singapore Airshow 2026; participated in the Women in Aviation panel talk on leadership and career pathways in aerospace.	
NTU Techignite	Jan 2024
Hands-on builds: water irrigation system, smart street lamp, hydraulic judo robot.	
NUS College of Design and Engineering (CDE) Art x Tech	Dec 2023
Co-built an Arduino-driven kinetic art installation.	
NUS CDE Summer Camp	Jun 2023
Led the “Cool House” build, applying convection cooling and frictional energy harvesting.	
Hey You Got Mail (HYGM)	2021 – 2023
Card-making and senior outreach; led a Crochet Session; 60+ volunteer hours.	

SECONDARY SCHOOL HIGHLIGHTS

Netball, Evergreen Secondary School	2019 – 2022
Publicity Manager (2021); School Representative, C & B Division Girls, North Zone.	

ACHIEVEMENTS & AWARDS

National Math Olympiad of Singapore (NMOS), Bronze	2017
44th National Lifesaving Championship & Singapore Life Saving Society (SLSS) Youth Lifesaving Championships, School Representative	2024

MICRO & ALTERNATIVE CREDENTIALS

Certificate of completion of Unmanned Aircraft Basic Training	2025
Level 1 Peer Supporter, Republic Polytechnic	2024
Civil Aviation Authority of Singapore (CAAS) Class 2 Medical Certificate (SYFC pilot training)	2024
AustSwim Teacher Licence	2022

REFERENCES

Available upon request.